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From Wicked Problem to Politically Managed Agenda: Evaluating the Efficiency–Equity Dilemma in Taiwan's *Nuclear-Free Homeland* Policy

1. **Introduction**

"Nuclear-Free Homeland" by 2025 is a contested policy that has been at the center of partisan politics in Taiwan. Policymaking in Taiwan, a relatively new representative democratic republic, mainly concerns the agenda set by those in control of the Executive power, the president, and the majority party in the legislative assembly. As the political culture is comprised of two dominating parties, the Democratic Progressive Party (DPP) and Kuomintang (KMT), the rise and fall of nuclear energy in Taiwan also outlined the tension between the two competing forces (Huang & Chen, 2021). Nuclear energy policy is often seen as a contested "wicked problem" navigating a pot-Fukushima East Asia. We would first explore its wickedness and find out that the case in Taiwan may not be as wicked and contested. However, when analyzing the policymaking process in the state's effort to phase out nuclear power, it is clear that the energy policy is highly complex. While first arises as a reaction to Fukushima and the social movement led by the people, supported by the DPP, the elected DPP showed a minimal effort to uphold the deliberative policymaking process—making the "Nuclear-Free Homeland" policy increasingly politicized. While the opposing discourse advocated against phasing out nuclear power by winning a referendum, the DPP government still proceeded with the original agenda. This paper argues that although the solution of implementing to phase out nuclear energy in Taiwan seems to be offering an excellent solution for concerns on nuclear safety, the process of policymaking is entirely top-down with minimal effort for a participatory opportunity—pointing to reasons like politicization and partisan politics. We will first outline the theoretical frameworks that help us understand wicked problems and policy analysis. Then, the paper will apply these frameworks to trace the evolution of Taiwan's nuclear energy policy—from its grassroots beginnings to the eventual top-

down implementation by the DPP—demonstrating how competing interests and a polarized political environment have shaped this contested agenda.

2. Theoretical Framework

To understand how Taiwan's nuclear energy policy is formulated, it is essential to begin with the concept of wicked problems. Rittel and Webber (1973) first conceptualized wicked problems as "complex, intractable, open-ended, unpredictable" policy challenges that defy clear definitions and resist straightforward, technical solutions. In essence, wicked problems involve multiple stakeholders with divergent interests and values, and any proposed solution tends to generate unforeseen consequences. As a policy area, nuclear energy exhibits many hallmarks of a wicked problem. In the post-Fukushima era, Taiwan's geological similarity to Japan increased public anxiety. Although nuclear energy has been present and has taken up a large share of energy consumption, the Fukushima disaster pushed this policy issue to the forefront of the Taiwanese public. Great geological activity on the island led to more than 82 percent, in a nationwide survey, of worried citizens, pointing to issues such as radioactive waste disposal, cancer risk, and nuclear accidents (Ho et al., 2014). On the other hand, the issue is more complicated than simply closing the nuclear plant for its cost-effectiveness. Simply turning away from nuclear power would mean switching to coal, natural gas, and fossil fuel, which are also environmentally costly (Yang et al., 2022). There is no doubt that Rittel and Webber would consider nuclear energy in Taiwan as wicked instead of "tame," yet this binary distinction cannot yield much insight into the nature of the problems.

A more nuanced typology, by Alford and Head (2017), extends beyond the binary categorization of wicked versus tame. The extension is in two key dimensions, the problem itself and the actors involved: one vertical dimension reflecting the complexity and uncertainty of the

problem and another horizontal dimension concerning the stakeholder environment. The level of wickedness in Taiwan's case is in the middle of the vertical dimension—the risk of nuclear energy is highly uncertain, and the public anxiety about a potential disaster makes it hard to navigate toward a definitive answer. However, we know that the problem is the negative externalities of utilizing nuclear power plants, not entirely navigating without direction. Meanwhile, on the horizontal dimension, the stakeholder environment in Taiwan is characterized by highly polarized political interests—putting them at the very top. DPP and KMT are on opposite ends of the policy issue, which is a stance they often adopt underlying their fundamental differentiations in Taiwan's independence. With these two dimensions, Alford and Head would characterize this energy issue as a "politically turbulent problem." This partisan environment leads to significant knowledge fragmentation and interest differentiation. While scientific experts may advocate for a balanced assessment of nuclear risks versus economic efficiency, public debates are reframed in strictly partisan terms (Huang & Chen, 2021; Chen, 2021). This theoretical framework sets the stage for analyzing how the nuclear energy policy evolved in Taiwan.

1. **Policy Process and Political Dynamics**

Anti-nuclear Position

Anti-nuclear energy started as a social movement as early as the late 1990s, and the then DPP was already at the forefront of that issue. However, it was not until Fukushima in 2011 that the movement gained momentum, and the grassroots effort culminated with the DPP starting to rally for a referendum to halt the construction of the fourth nuclear power plant (Huang 2021). At this stage, policy issues are brought to the forefront deliberatively, reflecting what Torgerson (1986) would identify as the third face of policy analysis—with a participatory focus. At that time, the pro-nuclear representation of the KMT was in power and faced grassroots efforts alongside the

DPP, and they held a referendum that contributed to the halt of the fourth power plant. Torgerson (1986) sees "engaging affected communities in both the design and implementation of policies" as a critical element. In this case, the momentum of the social movement also included the communities where nuclear waste got deposited in their neighborhood without procedural justice, and those who lived near the nuclear plants (Huang et al. 2021). Through Ensirnk's structure of the policy process, we can also identify this case as how policymaking is a discourse. In the argument, discourse focused on bringing the "advocacy coalitions" into conversation, leading to a plausible solution to the current problem. Other than halting the construction of a new power plant, this movement also gave birth to DPP's claim to bring forth a "Nuclear-Free Homeland." Broad public support for anti-nuclear efforts popularized their policy platform, which won DPP an overwhelming majority in the legislative assembly and presidential election in 2016—with a mission to accomplish this agenda.

Pro-nuclear Position

Including KMT, pro-nuclear power often adopts the rhetoric of technocratic and efficiency approach. Through quantitative analysis by Chen (2021) and Bergero et al. (2021), we must acknowledge that nuclear power can and has already contributed cost-effective and stable energy supply. Another central argument presented by the pro-nuclear group and analysis is the term "go green with nuclear," claiming that nuclear energy minimizes greenhouse gas emissions compared to other forms of energy production like coal (Woon 2021). While nuclear power plants also cause negative externalities in their surrounding area and produce toxic waste, their ability to reduce emissions on a grand scale is backed by scientific analysis (Chen 2021). With efficiency and low pollution as the center of the argument, abolishing nuclear entirely would mean that the government has to take on the cost of coal-burning pollution and less cost-effective means of

energy production—at least while developing alternatives. Forester outlines the debate between efficiency and equity in his claim on bounded rationality. "While the rational comprehensive formulation may reach for the sky, boundedly rational and incremental formulations may tie our hands," Forester argues that policy cannot operate solely through unbounded rationality. In this case, we would adopt nuclear energy as the most efficient solution and store the waste in the least populated area to minimize cost. However, with risk, multiple actors with varying interests, and public discontent bounding rationality, we must seek a balance between efficiency and equity. After examining the two perspectives through the policy analysis lens, we can shift towards examining how the policy process was brought out after the DPP took hold of the government.

Policy Process

After the general election of 2016, President Tsai solidified DPP's promise to follow the Nuclear-Free Homeland agenda and set 2025 as the year of Taiwan leaving nuclear power entirely. The agenda and the target electricity mix (increase in renewable energy (RE) and gas) are announced five days after the inauguration without any public participation and expert consultation prior (Gao, Yeh, and Chen 2022). The procedural flaw created backlash criticizing this top-down policymaking, and the DPP government held a public consultation a year later to salvage the mistake. Gao, Yeh, and Chen (2022) pointed out that in these consultations, the agenda of adopting a 15 percent increase in RE and leaving nuclear power entirely became an uncontested area of the Tsai administration—even after expressing dissent. The detail of the agenda was not changed after the seemingly performative participatory element. Huang and Chen (2021) claim that various approaches, legislative and executive, reflect a "decide-announce-defend" (DAD) approach that marginalizes public deliberation—especially civil society actors. By framing the nuclear phase-out as environmentally just, procedures that entail the change and the details are the opposite of

transparency and entirely block off from public consultation. The Tsai administration formed an ad-hoc institution called the "Office of Energy and Carbon Reduction." They bypassed institutional checks in assessing and assisting the energy transition in Taiwan. After the legislative majority DPP amended the Electricity Act in 2017, the government struggled to lift the development of RE. The amendment only liberalized electricity while officially mandating the phase-out of nuclear energy by 2025 (Gao et al. 2018). This DAD approach failed to provide Taiwan with a stable energy supply, increasing reliance on coal plants that are unable to control greenhouse emissions while causing air pollution and major power outages that created public discontent (Yang et al.).

The policy process is marked with little expert consultation, especially those who are not affiliated with the cabinet of the Tsai administration. While surveying scholars, Huang and Chen (2021) found that they feel disrespected when their professional opinion on nuclear phase-out is not considered entirely. "'Nuclear-free homeland' is a simple term – an energy mix of 50% natural gas, 30% coal, and 20% renewable energy, as I know, is just a campaign slogan lacking any robust theoretical support or scientific estimation for supporting the argument!" As Page (2010) noted, scientific expert consultation would lead to a certain degree of credibility. However, the DPP's top-down effort in constructing an energy transition without participatory measures and expert consultation negatively influenced the credibility of the Tsai Administration. KMT and pro-nuclear groups felt the public discontent built up and seized the opportunity. With the DPP controlling the legislative and executive branches, only a referendum would be feasible to signal public resistance to the current energy transition structure. The referendum held in 2018 asked the citizens whether they wanted to retract a critical bill in the recently amended Electricity Act to retract the "being nuclear-free by 2025" clause. While it passed at more than 59 percent of the

votes, the DPP downplayed the result of this deliberation (Huang et al. 2021). While all plants are on track to decommission by 2025, the Tsai Administration simply went forth with the agenda without considering nuclear as an option (Huang & Chen, 2021). Abolishing the clause vote by the referendum did not threaten DPP's agenda.

3. **Back to Theoretical**

Alford and Head (2017)outlined authoritative strategies as one of the ways to deal with wicked problems in literature, and this is exactly how the DPP is now taking upon this issue. Shifting from a more deliberative approach in developing the anti-nuclear campaign to now an entirely top-down approach that Torgerson (1986) would point to as the second face in policy analysis—when politics took over knowledge. Although one can argue that in the effort to phase out nuclear entirely, bounded rationality may lead to policymakers only being able to "satisfy" due to limited information, time, and cognitive capacity. However, the approach brought forth by the DPP entirely disregards citizen participatory and expert consultation. What initially was framed as an environmentally and socially just policy goal is now scrutinized mainly due to its policy process, which is highly politicized and partisan. As Craft and Halligan (2020) see in Westminster's counterparts, "attempts to increase partisan-political policy inputs" are the trend that is evident in Taiwan as polarization stretches between DPP and KMT.

After walking through the policy process of Taiwan's Nuclear energy policy issue, Nuclear-Free Homeland, the DPP's energy policy, is, in fact, politically reactive rather than scientifically guided. Though the wicked problem technically has been provided with a remedy, voices of contestation further prove that the issue is still complex. DPP's long political ties to the anti-nuclear agenda motivated the party to utilize extreme top-down means to reach policy goals, and the party was electorally punished partially, resulting from this approach. The case of nuclear energy in Taiwan

presented itself as the perfect textbook evidence to respect the participatory, deliberative process in policymaking, as presented by Torgerson and other scholars in policy analysis. Currently, KMT holds a legislative majority, and they have promised to propose amendments to the current version of the Electricity Act and put nuclear energy back as a policy option (Yahoo News 2025). With the relatively high level of air pollution from coal energy plants and unstable energy supply causing widespread outages in recent years, reconsidering nuclear energy seems more appealing than ever. Of course, with nuclear energy back in discussion, the repercussions are unavoidable—such as nuclear waste deposits. While no perfect policy would solve the wicked problem without any public discontent, deliberate has proven to be critical in facilitating the process.

4. **Conclusion**

In summary, nuclear energy as a policy issue meets many of the criteria of a wicked problem according to both the classical definition (Rittel & Webber, 1973) and the more nuanced typology proposed by Alford and Head (2017). In Taiwan, although the nuclear-free agenda is publicly portrayed as a participatory and deliberative process, the reality is that decision-making is dominated by elite actors who simplify the complexity into a binary policy choice. This reductionist approach fails to address the intricate trade-offs between economic efficiency and equitable risk distribution, ultimately reflecting a politicized process that sidelines the very dimensions of wickedness that require iterative, inclusive, and adaptive governance. Applying this theoretical framework to the Taiwanese case highlights the need for more contingent and context-sensitive approaches that integrate technical assessments with genuine stakeholders.

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